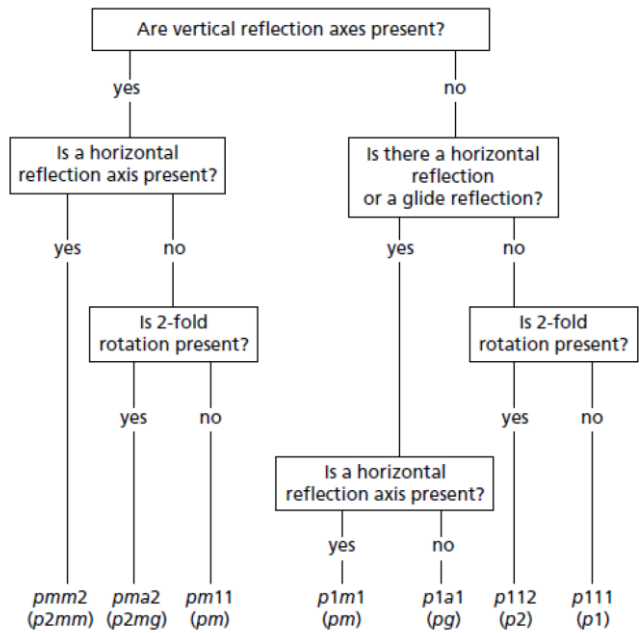


HW1 - Question 1



R R R R R R p1

$\frac{R}{B} \frac{R}{B} \frac{R}{B} \dots$ pg

$\frac{R}{B} \frac{R}{B} \frac{R}{B} \frac{R}{B}$ p2

| R | Я | R | Я | R | Я | pm

$\frac{R}{B} \frac{R}{B} \frac{R}{B} \frac{R}{B} \frac{R}{B}$ pm

$\frac{R}{B} \frac{R}{B} \frac{R}{B} \frac{R}{B}$ p2mg

$\frac{R}{B} \frac{R}{B} \frac{R}{B} \frac{R}{B} \frac{R}{B} \frac{R}{B}$ p2mm

p1 E

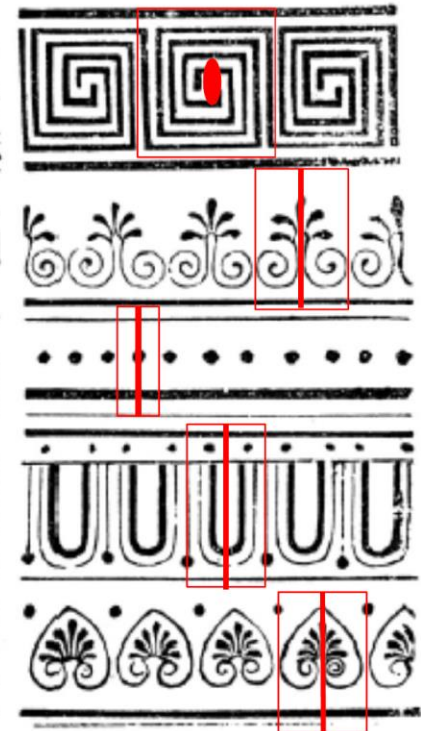
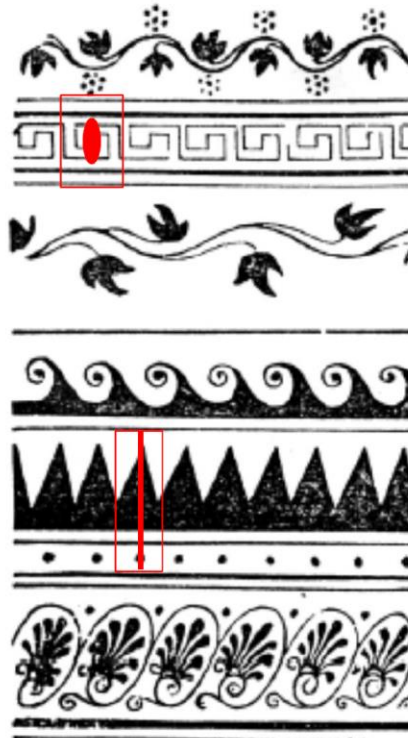
p2 F

p1 G

p1 H

pm I

p1 J



K p2

L pm

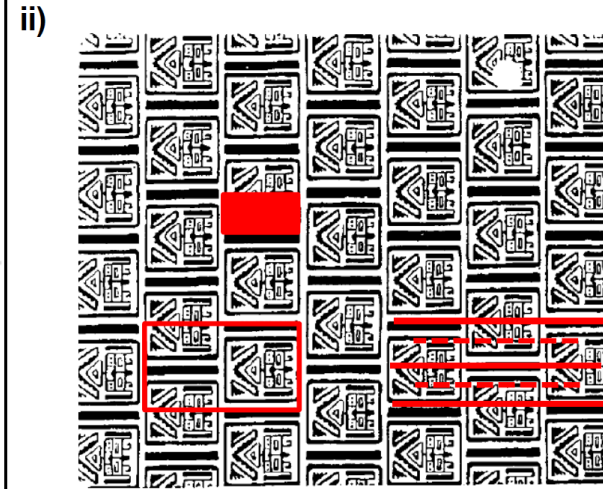
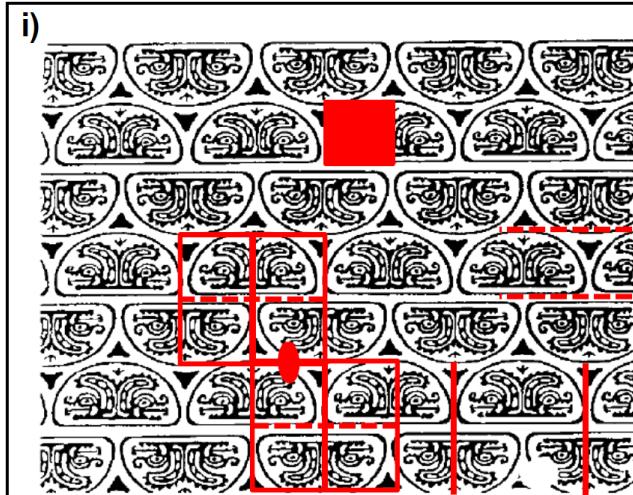
M pm

N pm

O pm

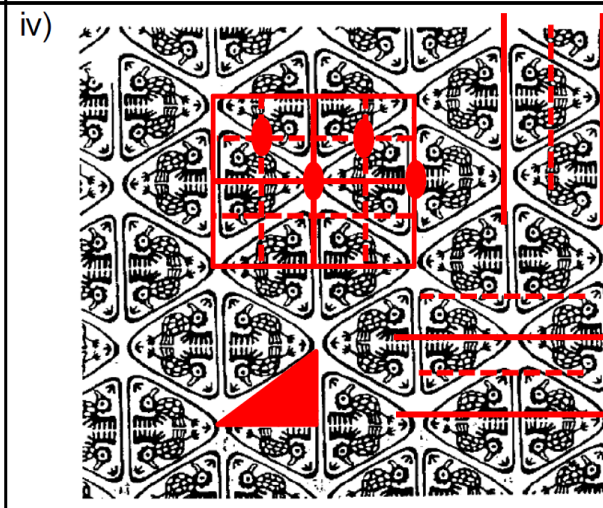
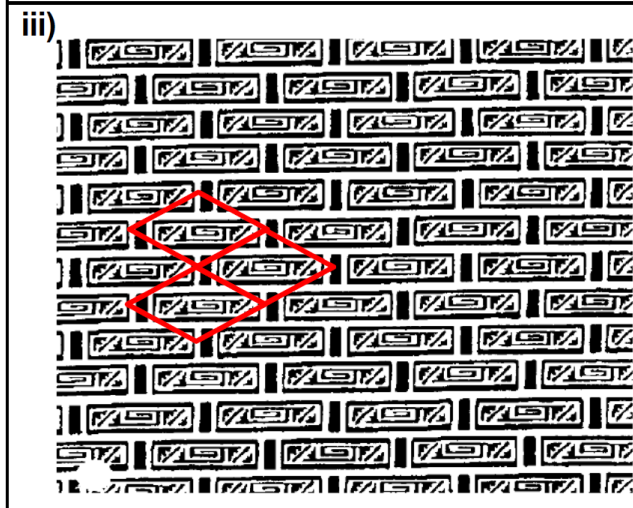
HW1 - Question 2

Primitive
rectangular
($p2mg$)



Centred
rectangular
(cm)

Primitive
oblique
($p1$)



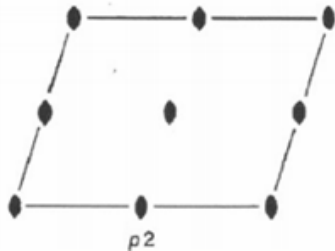
Centred
rectangular
($c2mm$)

HW1 - Question 3 a

Q3) Decoding Plane Groups (30 pts)

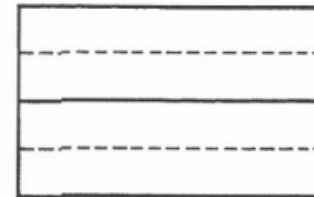
a) For each plane group below, identify the 2D Bravais lattice and symmetry elements associated to the following plane groups. Are there additional symmetries not explicitly mentioned in the plane group? (18 pts)

p2



Oblique
Diads

cm



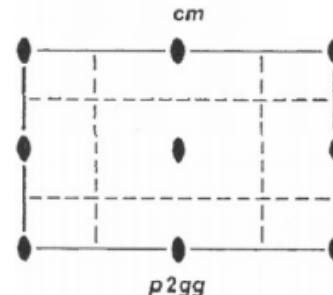
Rectangular C
Mirror and
glide lines

pg



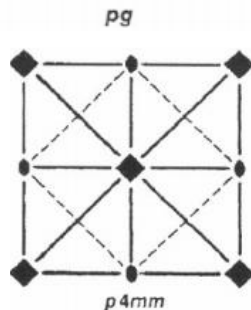
Rectangular P
Glide lines

p2gg



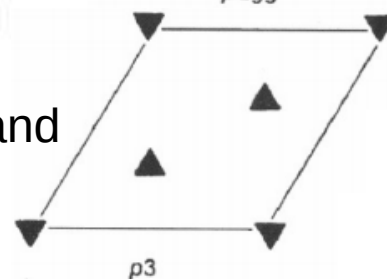
Rectangular P
Diads and
multiple glide
lines

p4mm



Square
Tetrads, Diads, and
multiple mirrors
Inherent glides

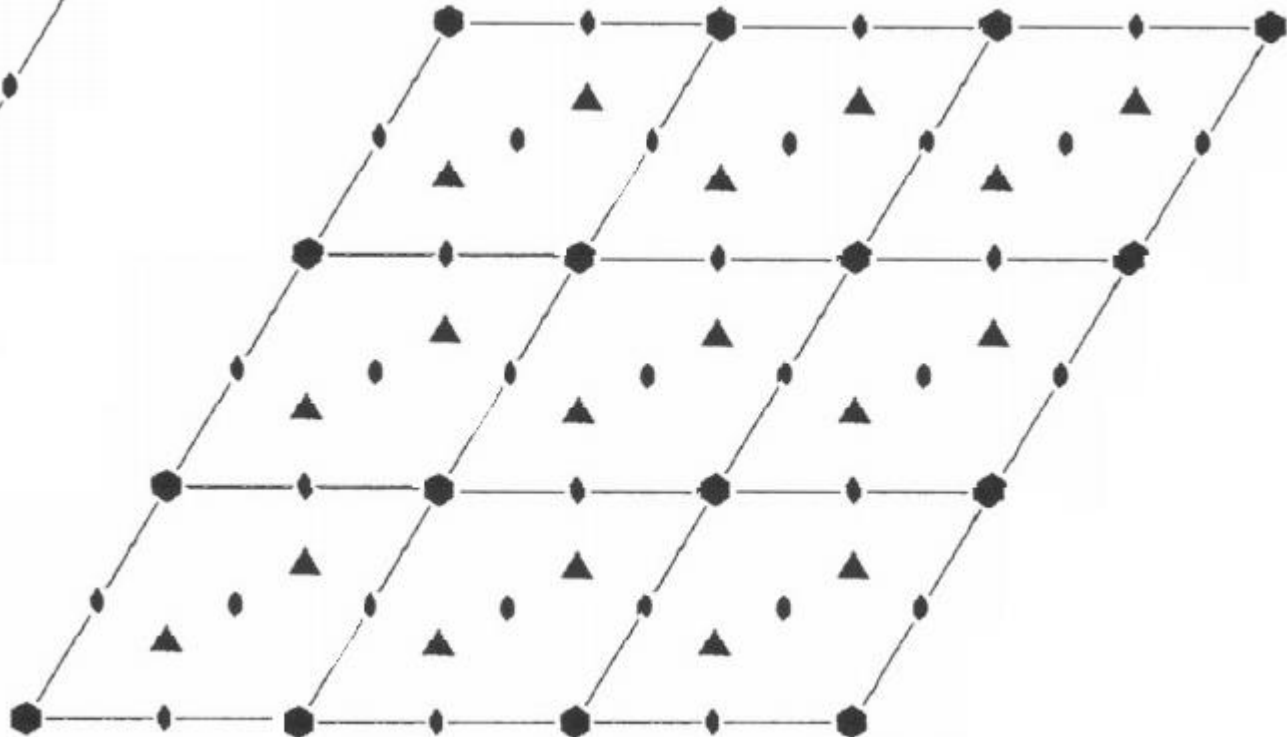
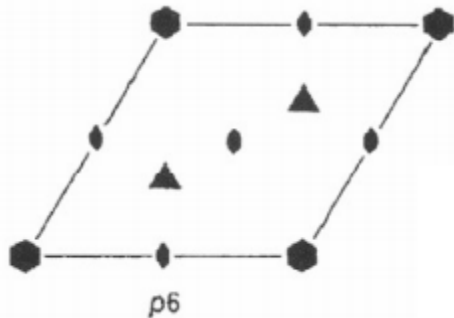
p3



Hexagonal
Triads

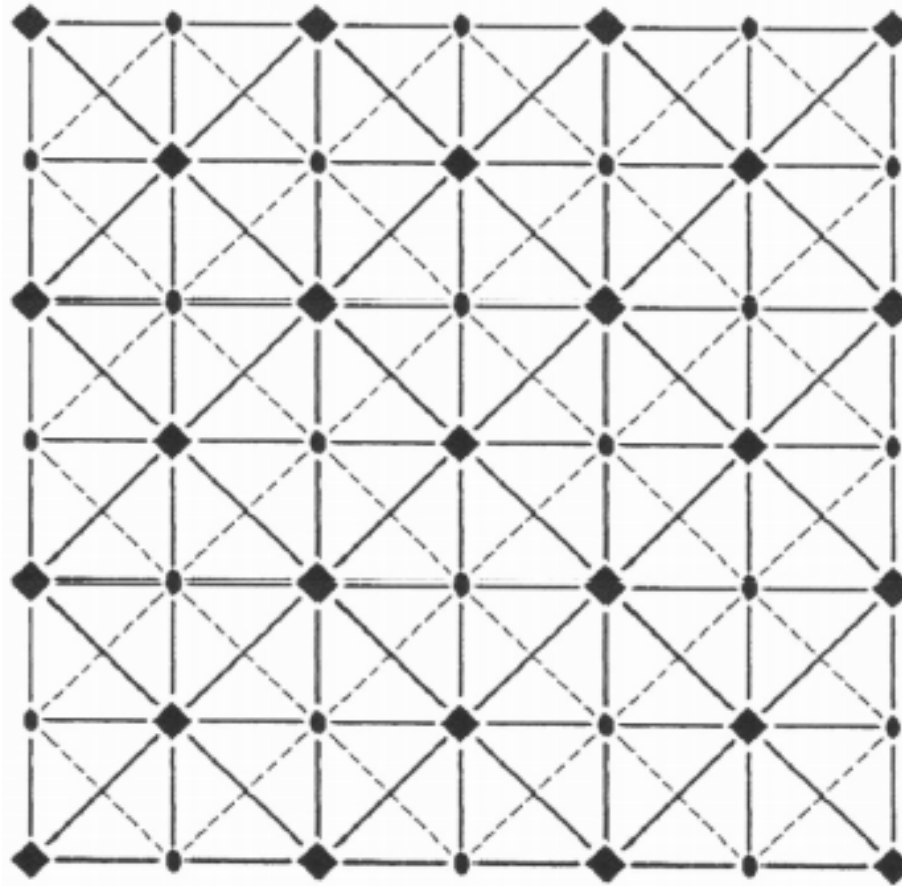
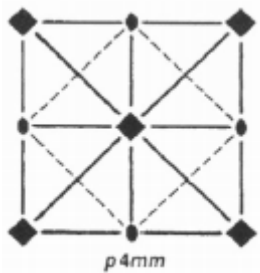
HW1 - Question 3 b

- b) Draw a p6 lattice and illustrate 9 unit cells in 2D. Place the essential symmetry at the corners of the unit cell and infer which additional symmetries are present in the 2D crystal which are not explicitly mentioned in the plane group. (6 pts)



HW1 - Question 3 c

- c) Draw a $p4mm$ lattice and illustrate 9 unit cells in 2D. Place the essential symmetries within the unit cell and infer which additional symmetries are present within the 2D crystal. (6 pts)



HW1 - Question 4 a

- a) 2D lattice: A point with coordinates (x,y) is rotated in-plane by a triad located at the origin. Calculate the coordinates of all the points generated by the triad. (15 pts)

$$R_z(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$\Theta = 120$ degrees & 240 degrees for triad

$$(x_1, y_1) = (x, y)$$

$$(x_2, y_2) = \begin{bmatrix} \cos(120) & -\sin(120) & 0 \\ \sin(120) & \cos(120) & 0 \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} x \\ y \\ 0 \end{bmatrix} = \begin{bmatrix} -\frac{1}{2}x - \frac{\sqrt{3}}{2}y \\ \frac{\sqrt{3}}{2}x - \frac{1}{2}y \\ 0 \end{bmatrix}$$

$$(x_3, y_3) = \begin{bmatrix} \cos(240) & -\sin(240) & 0 \\ \sin(240) & \cos(240) & 0 \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} x \\ y \\ 0 \end{bmatrix} = \begin{bmatrix} -\frac{1}{2}x + \frac{\sqrt{3}}{2}y \\ \frac{\sqrt{3}}{2}x + \frac{1}{2}y \\ 0 \end{bmatrix}$$

HW1 - Question 4 b

- b) 1D lattice: A point located at $(1/3, 1/6)$ is subject to a glide-reflection line parallel to the y-axis. The Glide-reflection intercepts the x-axis at $x = -1$. Calculate the coordinate of the new point. Assuming these two points form the basis, identify the unit cell, draw symmetry elements, and identify the line group. (10 pts)

Translate by $t/2$ in y direction

Reflect across $x = -1$

Define unit cell that captures both points

Symmetry element: Glide line

Type: pg

